



MANAGING ORGANIZATIONS

Beware Spurious Correlations

From the June 2015 Issue

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We all know the truism “Correlation doesn’t imply causation,” but when we see lines sloping together, bars rising together, or points on a scatterplot clustering, the data practically begs us to assign a reason. We want to believe one exists.

Statistically we can’t make that leap, however. Charts that show a close correlation are often relying on a visual parlor trick to imply a relationship. Tyler Vigen, a JD student at Harvard Law School and the author of *Spurious Correlations*, has made sport of this on his website, which charts farcical correlations—for example, [between U.S. per capita margarine consumption and the divorce rate in Maine](#).

Vigen has programmed his site so that anyone can find and chart absurd correlations in large data sets. We tried a few of our own and came up with these gems:

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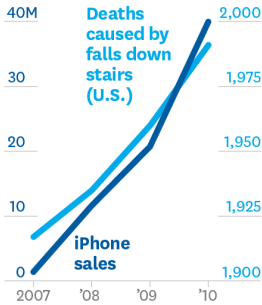
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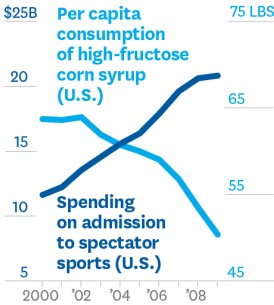
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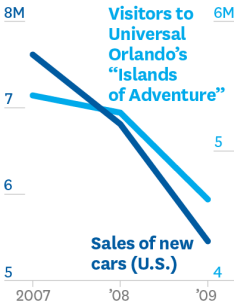
MORE IPHONES MEANS
MORE PEOPLE DIE FROM
FALLING DOWN STAIRS



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THE TEAM, AND
WE’LL LOSE WEIGHT



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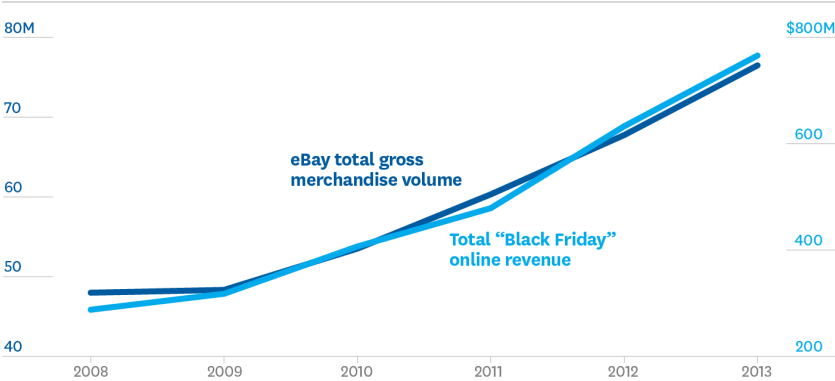
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Although it’s easy to spot and explain away absurd examples like these, you’re likely to encounter rigged but plausible charts in your daily work. Here are three types managers should watch out for:

Apples and Oranges Comparing Dissimilar Variables

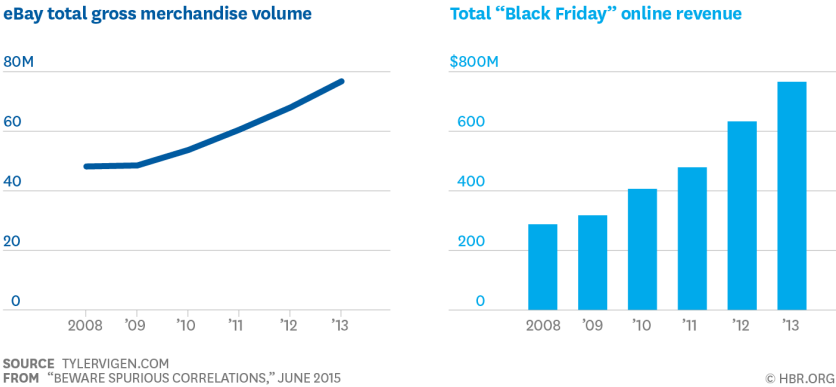
Y axis scales that measure different values may show similar curves that shouldn’t be paired. This becomes pernicious when the values appear to be related but aren’t.



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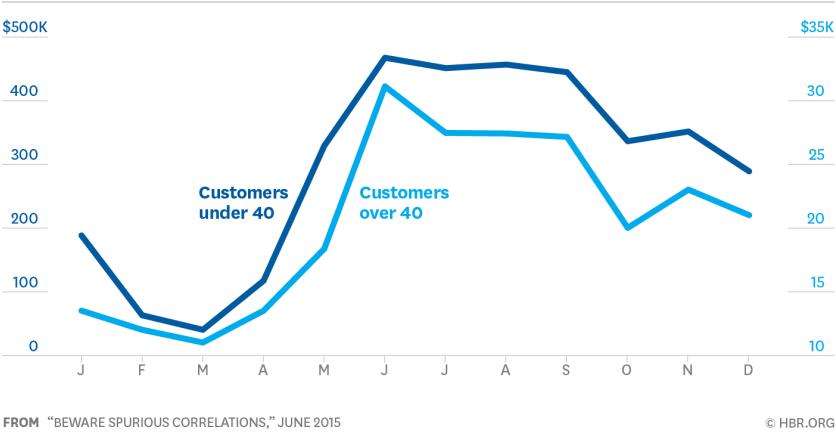
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It’s best to chart them separately.

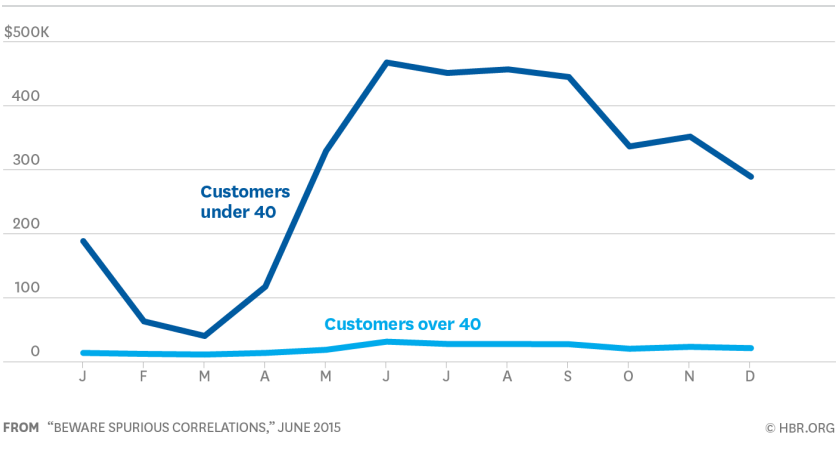


Skewed Scales Manipulating Ranges to Align Data

Even when Y axes measure the same category, changing the scales can alter the lines to suggest a correlation. These Y axes for RetailCo’s monthly revenue differ in range and proportional increase.

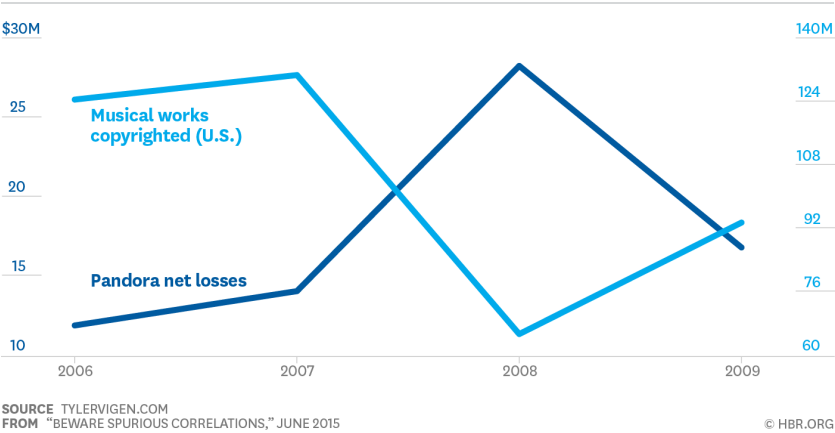


Eliminating the second axis shows how skewed this chart is.

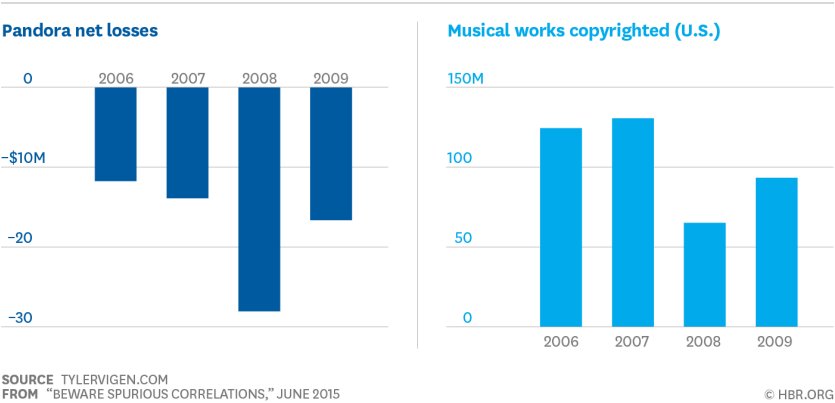


Ifs and Thens Implying Cause and Effect

Plotting unrelated data sets together can make it seem that changes in one variable are causing changes in the other.



We try to create a narrative—*If* Pandora loses less money, *then* more music is copyrighted—from what is probably a coincidence.



A version of this article appeared in the [June 2015](#) issue (pp.34–35) of *Harvard Business Review*.

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